25901152 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSECHE200 - POLYMER CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very short answer questions

Answer any **10** questions

Each question carries **2** marks

1. Define the chain initiation step in free radical polymerization and provide a chemical equation for it. [U] / [CO1]
2. What are shape-memory polymers? [K] / [CO1]
3. Give an example for a monomer which undergoes cationic polymerization explain why it undergoes this type of polymerization. [U] / [CO1]
4. What is atactic polymer? Give an exmple. [U] / [CO2]
5. Give two examples of polymers prepared by suspension polymerisation. [U] / [CO2]
6. Explain the effect of plasticizers on glass transition temperature. [U] / [CO2]
7. What structural change occurs during vulcanisation of rubber? [U] / [CO3]
8. Identify the polymers from the monomers: (i) caprolactum and (ii) hexamethylene diamine + adipic acid [U] / [CO3]
9. Identify the monomers used to make PET polymers. [K] / [CO3]

10. Describe the term polymer reagent with reference to polymer support. [U] / [CO4]

11. How can temperature sensitive polymers be utilized to achieve targeted drug delivery? [A] / [CO4]

12. Compare the electrical conductivity of polyacetylene with other conducting polymers. [A] / [CO4]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

13. Comment on the importance of DNA as a polymer. [U] / [CO1]

14. Explain the hydrolytic ring-opening polymerisation mechanism of Nylon-6. [K] / [CO1]

15. Compare solution polymerisation and emulsion polymerisation. [U] / [CO2]

16. Explain the factors influencing the crytallinility and amorphous characters of polymers? [A] / [CO2]

17. Describe the structure and properties of Urea–formaldehyde resin [K] / [CO3]

18. Give the structure of polystyrene and discuss a few of its applications. [U] / [CO3]

19. Describe the application of Nomex in the field of protective clothing. [U] / [CO4]

20. Discuss the applications of poly vinyl carbazole (PVK) in photoconducting polymers. [A] / [CO4]

[5x6 = 30]

Part C

Short Essay Questions

Answer any **2** questions

Each question carries **10** marks

21. Explain in detail the classification of polymers based on origin and structure. [U] / [CO1]

22. Explain in detail the practical significance of polymer molecular weight? [An] / [CO2]

23. Describe the properties and applications of PMMA. [U] / [CO3]

24. Discuss the structure-property relationship, and applications of biodegradable polyurethanes. [U] / [CO4]

[10x2 = 20]